STAT 3400 - Homework 2

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Problem 1

You will now think of some real-life applications for statistical learning. Explain your answers completely.

Part A

Describe three real-life applications in which *classification* might be useful. Describe the response, as well as the predictors. Is the goal of each application inference or prediction?

Solution:

The first example of a real-world classification application is online shopping advertisements based on customer groups. In this case, the response variable is the customer group, and the predictors include past purchase history, browsing behavior, and demographic information. The goal of this application is inference, because companies are interested in understanding patterns and trends within each group so they can implement different discount strategies or making approaches for different types of customers. The second example is fraud detection in the financial industry. Here, the response variable is whether a transaction is fradulent or not, and the predictors include transaction amount, location, time of transaction, and historical spending behavior. This is a classification problem where the primary goal is prediction, since financial institutiions need to quickly and accurately classify new transactions to prevent fraud. The last example is email spam detection based on the content of emails. In this application, the response variable is whether an email is spam or not spam, and the predictors include word frequency, presence of links, sender information, and email length. This is mainly a prediction problem, as the objective is to automatically classify incoming emails correctly.

Part B

Describe three real-life applications in which *regression* might be useful. Describe the response, as well as the predictors. Is the goal of each application inference or prediction?

Solution:

One well-known real-world application of regression is predicting the performance of a soccer team over a season. The response variable could be the total points or win percentage, while the predictors include past performance, team budget, player salaries, and injury information. The goal of this application is prediction, since the focus is on forecasting future performance. The second example is estimating real estate prices. In this case, the response variable is the house price, and the predictors include location, size of the house, number of bedrooms, lot size, and age of the property. This application involves both inference and prediction, because regression is used to predict prices as well as to understand how each factor affects the value of a property. The final example is predicting the S&P 500 closing price using historical data. The response variable is the closing price of the index, and the predictors include previous prices, trading volume, and the economic indicators. This is primarily a prediction problem, often approached using time-series models such as ARIMA or SARIMA to forecast future market behavior.

Problem 2

This problem involves the `Auto` data set from the `ISLR2` library. After loading the library, information about the variables included in the data set can be found by running `?Auto` in the console.

```
library(ISLR2)
data(Auto)
?Auto
```

Part A

Suppose that we wish to predict gas mileage (`mpg`) on the basis of the other variables. Which of the predictors are quantitative, and which are qualitative? For each quantitative predictor, provide the range, mean, and standard deviation. For each qualitative predictor provide the mode.

Solution:

- Qualitative predictors: `culinders`, `displacement`, `horsepower`, `weight`, `acceleration`, and `year`.
- Qualitative predictors: `origin`, `name`.

```
Auto <- na.omit(Auto)
cat("Cylinders:\n")
```

```
## Cylinders:
```

```
cat("  Range:", range(Auto$cylinders), "\n")
```

```
##   Range: 3 8
```

```
cat("  Mean:", mean(Auto$cylinders), "\n")
```

```
##   Mean: 5.471939
```

```
cat("  Standard Deviation:", sd(Auto$cylinders), "\n\n")
```

```
##   Standard Deviation: 1.705783
```

```
cat("Displacement:\n")
```

```
## Displacement:
```

```
cat("  Range:", range(Auto$displacement), "\n")
```

```
##   Range: 68 455
```

```
cat("  Mean:", mean(Auto$displacement), "\n")
```

```
##   Mean: 194.412
```

```
cat("  Standard Deviation:", sd(Auto$displacement), "\n")
```

```
##   Standard Deviation: 104.644
```

```
cat("Horsepower: \n")
```

```
## Horsepower:
```

```
cat("  Range:", range(Auto$horsepower), "\n")
```

```
##   Range: 46 230
```

```
cat("  Mean:", mean(Auto$horsepower), "\n")
```

```
##   Mean: 104.4694
```

```r
cat("  Standard Deviation:", sd(Auto$horsepower), "\n")
```

```
##   Standard Deviation: 38.49116
```

```r
cat("Wegiht: \n")
```

```
## Wegiht:
```

```r
cat("  Range:", range(Auto$weight), "\n")
```

```
##   Range: 1613 5140
```

```r
cat("  Mean:", mean(Auto$weight), "\n")
```

```
##   Mean: 2977.584
```

```r
cat("  Standard Deviation:", sd(Auto$weight), "\n")
```

```
##   Standard Deviation: 849.4026
```

```r
cat("Acceleration: \n")
```

```
## Acceleration:
```

```r
cat("  Range:", range(Auto$acceleration), "\n")
```

```
##   Range: 8 24.8
```

```r
cat("  Mean:", mean(Auto$acceleration), "\n")
```

```
##   Mean: 15.54133
```

```r
cat("  Standard Deviation:", sd(Auto$acceleration), "\n")
```

```
##   Standard Deviation: 2.758864
```

```r
cat("Year: \n")
```

```
## Year:
```

```r
cat("  Range:", range(Auto$year), "\n")
```

```
##   Range: 70 82
```

```r
cat("  Mean:", mean(Auto$year), "\n")
```

```
##   Mean: 75.97959
```

```r
cat("  Standard Deviation:", sd(Auto$year), "\n")
```

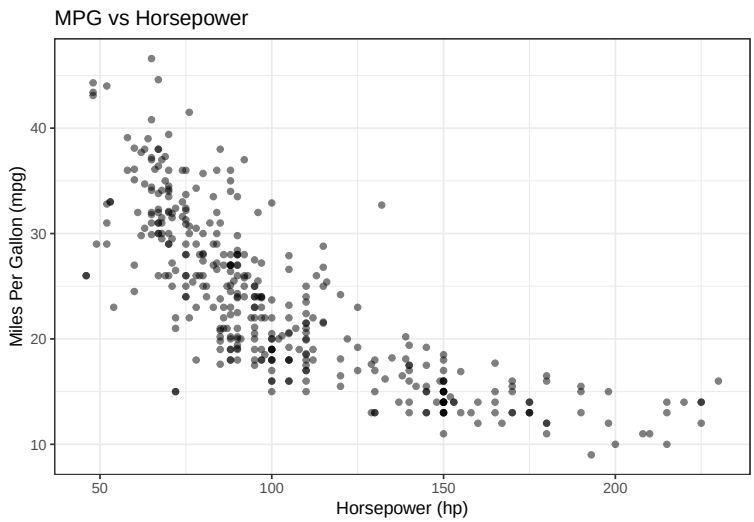
```
##   Standard Deviation: 3.683737
```

Part B

Investigate the predictors graphically, using scatter plots or other tools of your choice. Create *three* plots that depict the association between gas mileage and one of the predictors. Be sure to include titles, axis labels, and units on your plots. Comment on your findings from each plot. Specifically, do any of the plots suggest a strong association between `mpg` and the other variables?

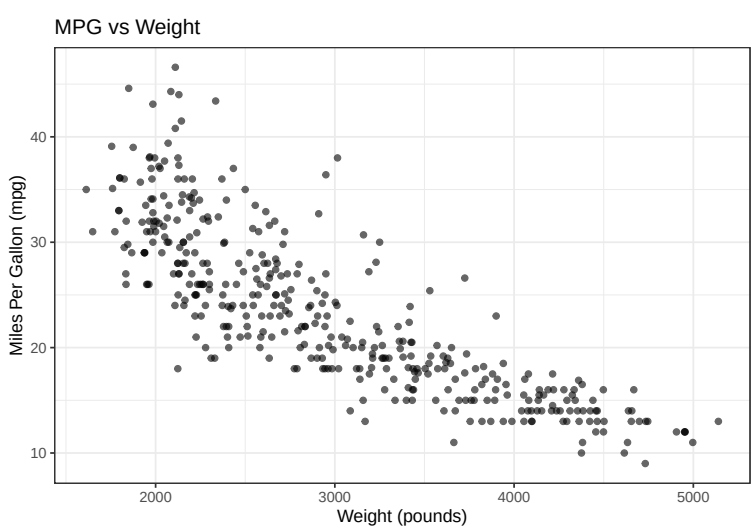
```r
library(ggplot2)

ggplot(Auto, aes(x = horsepower, y = mpg)) +
```

```r
  geom_point(alpha = 0.5) +
  labs(title = "MPG vs Horsepower",
       x = "Horsepower (hp)",
       y = "Miles Per Gallon (mpg)") +
  theme_bw()
```



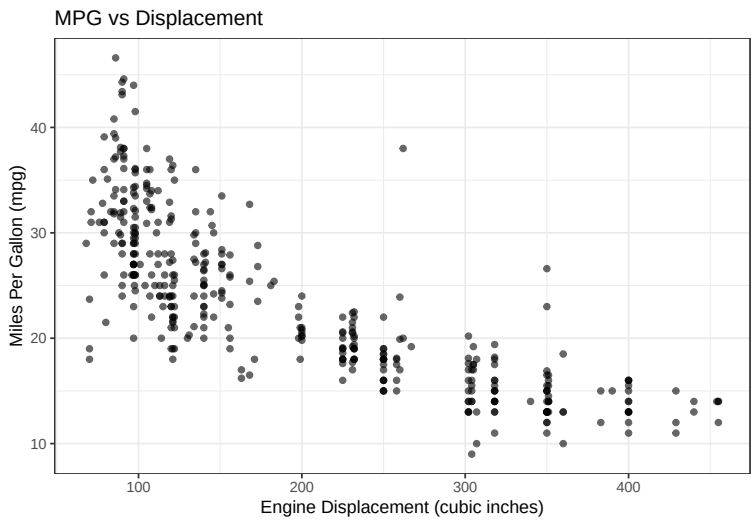
Comments: This plot shows the relationship between mpg and horsepower. There is a clear negative association between horsepower and mpg, indicating that cars with higher horsepower tend to have lower gas mileage. The relationship appears relatively strong and approximately linear or maybe quant.

```r
ggplot(Auto, aes(x = weight, y = mpg)) +
  geom_point(alpha = 0.6) +
  labs(
    title = "MPG vs Weight",
    x = "Weight (pounds)",
    y = "Miles Per Gallon (mpg)"
  ) +
  theme_bw()
```



Comments: This plot displays vehicle weight. This plot shows a very strong negative association, with heavier cars consistently achieving lower gas mileage. Among the three predictors considered, weight appears to have the strongest association with mpg.

```
ggplot(Auto, aes(x = displacement, y = mpg)) +
  geom_point(alpha = 0.6) +
  labs(
    title = "MPG vs Displacement",
    x = "Engine Displacement (cubic inches)",
    y = "Miles Per Gallon (mpg)"
  ) +
  theme_bw()
```



Comments: The plot shows the relationship between mpg and engine displacement. There is also a negative association between displacement and mpg, meaning that cars with larger engines tend to be less fuel efficient. Although the relationship is clear, it appears slightly more dispersed than the

Overall, all three plots suggest strong negative associations between mpg and the predictors examined, with vehicle weight showing the strongest relationship.

Problem 3

Referring again to the `Auto` data, suppose we wish to assess the accuracy of a model to predict a vehicle's gas mileage (`mpg`). Randomly split the data into 75% training and 25% testing using `set.seed(720)`. Then answer the following questions.

```
set.seed(720)

rows <- sample(x = nrow(Auto), size = floor(0.75 * nrow(Auto)), replace = FALSE)

training <- Auto[rows, ]
testing <- Auto[-rows, ]
```

Part A

If we always predict the mean gas mileage from the training set, what is the root mean squared error (RMSE) against the testing set? Does this seem very accurate? Why or why not?

```r
mpg_avg <- mean(training$mpg)
RMSE <- sqrt(mean((testing$mpg - mpg_avg)^2))
RMSE
```

```
## [1] 7.217143
```

Comments: In Part A, the null model predicts gas mileage using the mean mpg from the training data for all observations in the testing set. The resulting RNSE is approximately 7.22. This level of error is relatively large compared to the typical range of mpg values, indicating that the null model does not provide very accurate predictions. Since this approach ignores all predictor information, it lacks flexibility and can't capture variation in mpg across different vehicles.

Part B

Now suppose we train a simple linear regression model to predict gas mileage based on the vehicle's horsepower. What is the root mean squared error (RMSE) against the testing set? How does this accuracy compare to the null model from Part A?

```r
lm_model <- lm(mpg ~ horsepower, data = training)
predictions <- predict(lm_model, newdata = testing)
lm_RMSE <- sqrt(mean((testing$mpg - predictions)^2))
lm_RMSE
```

```
## [1] 4.524103
```

Comments: In Part B, a linear regression model was used to predict mpg based on horsepower. The RMSE for this model is approximately 4.52, which is substantially lower than the RMSE from the null model in Part A. This improvement suggests that horsepower contains meaningful information fro predicting mpg. As observed in the plot from Problem 2, there is a strong negative and approximately linear relationship between `mpg` and `horsepower`, which helps explain the improved predictive accuracy of the linear regression model.

Problem 4

This problem involves the `Boston` data set from the `ISLR2` library. After loading the library, information about the variables included in the data set can be found by running `?Boston` in the console.

```r
data(Boston)
?Boston
```

Part A

How many rows are in this data set? How many columns? What does each row (observation) and column (variable) represent? What is the original source of this data?

Solution: A data contains 506 rows and 13 variables, and this dataset was obtained from the Boston dataset that is part of the MASS library.

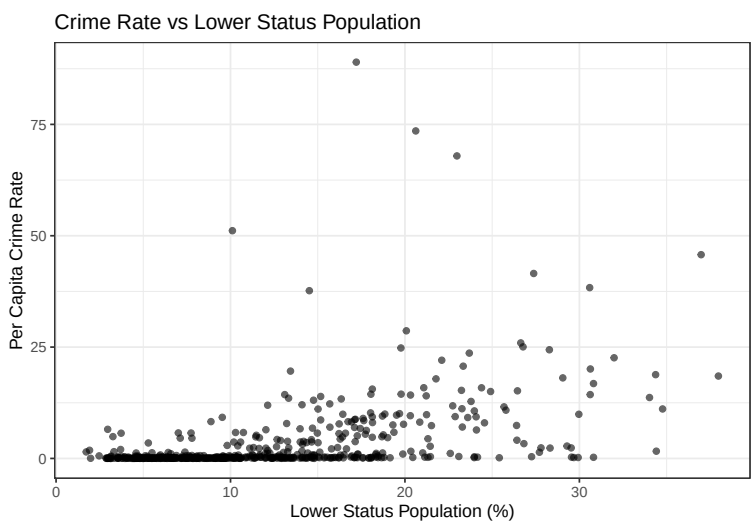
- `crim`: per capita crime rate by town.
- `zn`: proportion of residential land zoned for lots over 25,000 sq.ft.
- `indus`: proportion of non-retail business acres per town.
- `chas`: Charles River dummy variable.
- `nox`: nitrogen oxides concentration.

- `rm`: average number of rooms per dwelling.
- `age`: proportion of owner-occupied units built prior to 1940.
- `dis`: weighted mean of distances to five Boston employment centres.
- `rad`: index of accessibility to radial highways.
- `tax`: full-value property-tax rate per $10,000.
- `ptratio`: pupil-teacher ratio by town.
- `lstat`: lower status of the population (percent).
- `medv`: median value of owner-occupied homes in $1000s.

Part B

Suppose that we wish to predict per capita crime rate (`crim`) based on the other variables. Create *three* plots that depict the association between crime rate and one of the predictors. Be sure to include titles, axis labels, and units on your plots. Comment on your findings from each plot. Specifically, do any of the plots suggest a strong association between `crim` and the other variables?

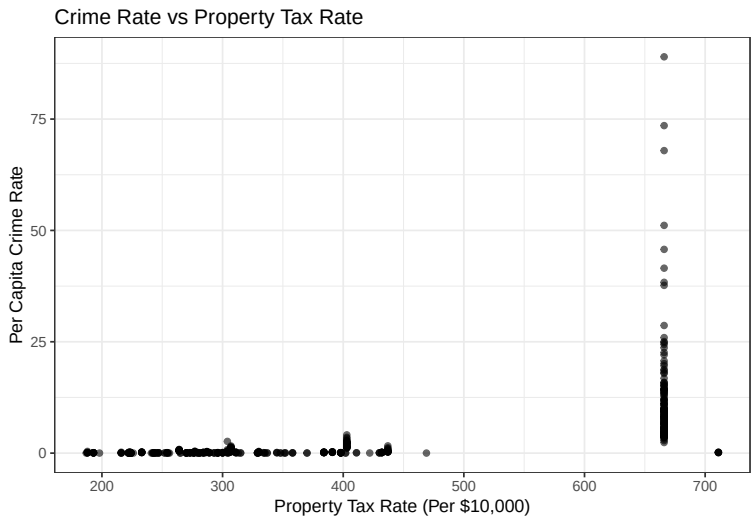
```
ggplot(data = Boston, aes(x = lstat, y = crim)) +
  geom_point(alpha= 0.6) +
  labs(title = "Crime Rate vs Lower Status Population",
       x = "Lower Status Population (%)",
       y = "Per Capita Crime Rate") +
  theme_bw()
```



Comments: This plot shows the relationship between per capita crime rate and the percentage of the lower status population. There is a moderate positive association between `lstat` and `crim`, indicating that towns with a higher proportion of lower status residents tend to have higher crime rates. While many towns with low values of `lstat` have very low crime rates, crime becomes more variable and generally higher as `lstat` increases. The presence of several extreme values at higher `lstat` levels suggest potential outliers, but the overall upward trend is evident. This plot suggests a strong association between lower status population and per capita crime rate.

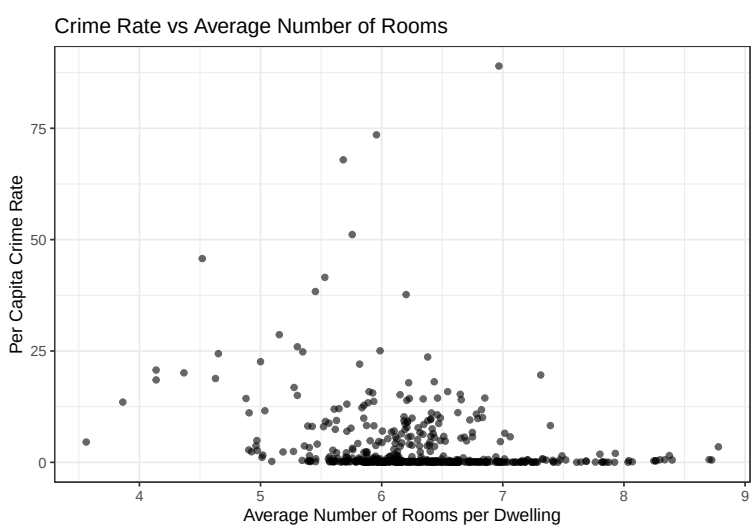
```
ggplot(data = Boston, aes(x = tax, y = crim)) +
  geom_point(alpha = 0.6) +
  labs(title = "Crime Rate vs Property Tax Rate",
```

```
       x = "Property Tax Rate (Per $10,000)",
       y = "Per Capita Crime Rate") +
  theme_bw()
```



Comments: This plot shows the relationship between per cpita crime rate and the property tax rate. There is a slightly positive association between `tax` and `crim` with towns that have higher property tax rates generally exhibitning higher crime rates. However, the relationship appears weaker and more dispersed compared to the association between crime rate and lower status population. Much of the variability in crime rate occurs at higer tax values, and the clustering of observations suggests that property tax rate alone may not fully explain differences in crime rates. Overall, while a positive trend is present, the association between `tax` and `crim` is moderate rather than strong.

```
ggplot(Boston, aes(x = rm, y = crim)) +
  geom_point(alpha = 0.6) +
  labs(
    title = "Crime Rate vs Average Number of Rooms",
    x = "Average Number of Rooms per Dwelling",
    y = "Per Capita Crime Rate"
  ) +
  theme_bw()
```



Comments: This plot shows the relationship between per capita crime rate and the average number of rooms per dwelling. There is a weak negative association, indicating that towns with larger average home sizes tend to have lower crime rates. Most observations with higher values of `rm` are associated with very low crime rates, while higher crime rates are common among towns with fewer rooms per dwelling. Although there is some variability and a few outliers, the overall downward trend suggest that a moderate negative association between `rm` and `crim`.